Time Series Econometrics for Finance

Assignment 2 — Solutions

A. Empirical Exercise: Unit-Root Test on Log Real GNP

1) Series Selection and Plot

The series chosen is **Real GNP (RGNP)** from the Nelson-Plosser dataset. After removing the leading zero observations (1860–1908), the usable annual sample runs from **1909 to 1970** (T = 62 observations). The log transformation $y_t = \ln(RGNP_t)$ is applied as instructed.

The upper panel of Figure 1 shows a clear and persistent *upward trend* with no obvious tendency to revert to a fixed mean — a first visual indication of non-stationarity. The lower panel shows the first difference (annual growth rates), which fluctuates around zero with no trend, consistent with stationarity after differencing.

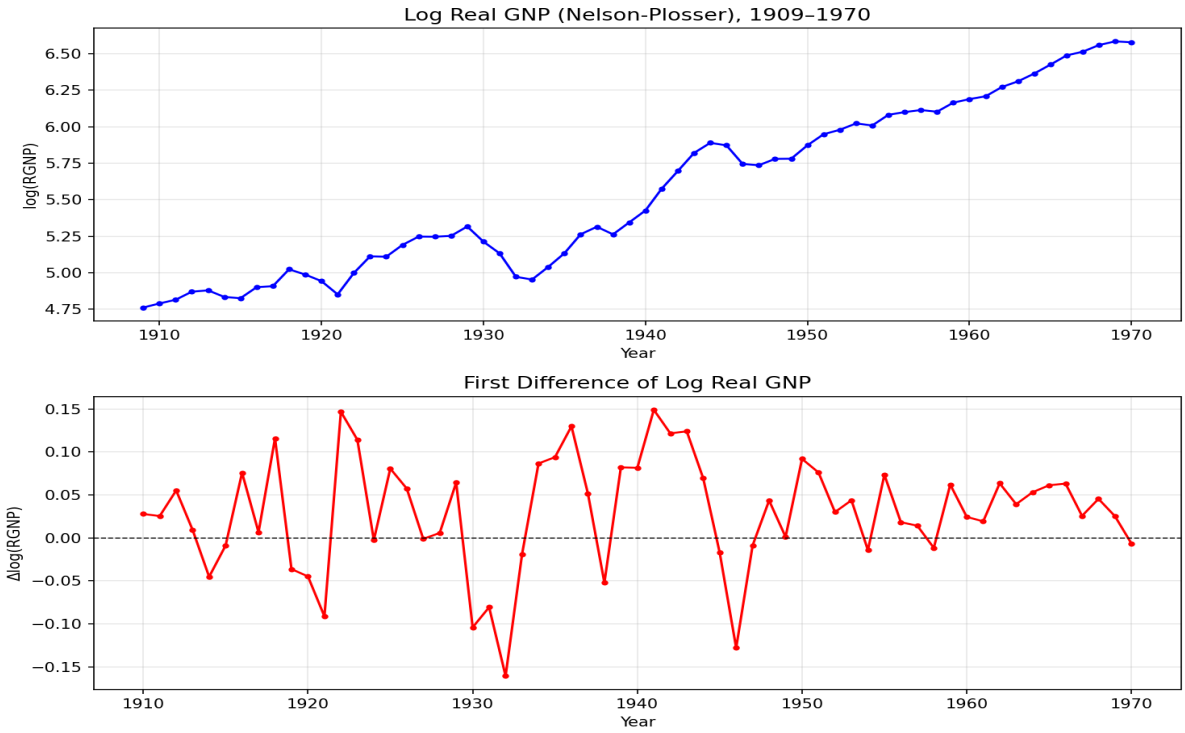

Figure 1. Log Real GNP (top) and its first difference (bottom), 1909–1970.

2) Data Preparation

Observations with zero values (years 1860–1908) are dropped, leaving 62 annual observations. The series is log-transformed: $y_t = \ln(RGNP_t)$. RGNP is not the Bond Yield series, so the log transformation applies.

3) Unit-Root Testing Format

We follow the **top-down approach** from the Lecture Notes (pp. 138–141), incorporating prior information from the visual inspection of the series:

Step 1. Since the log level clearly trends upward, we begin with the most general ADF specification — **Model 3** (constant + deterministic trend):

$$\Delta y_t = a_0 + a_2 t + \gamma y_{t-1} + b_1 \Delta y_{t-1} + \varepsilon_t$$

Step 2. Lag order selected by BIC over lags 0–4. BIC selects **p = 1**.

Step 3. The key test is the **joint F-test (φ_3)**, which tests H_0: $\gamma = 0$ *and* $a_2 = 0$ simultaneously (unit root with no deterministic trend). The restricted model under H_0 is:

$$\Delta y_t = a_0 + b_1 \Delta y_{t-1} + \varepsilon_t \quad \text{(random walk with drift)}$$

The F-statistic is compared to the Dickey-Fuller φ_3 critical values (5%: 6.49, 10%: 5.47). Standard t-table critical values do *not* apply under the unit-root null.

4) Results and Interpretation

Test	Statistic	CV 1%	CV 5%	CV 10%	Decision
ADF t-stat (ct, p=1)	-2.9939	-4.118	-3.486	-3.171	Fail to reject H_0
φ_3 F-stat (ct, p=1)	4.619	—	6.49	5.47	Fail to reject H_0
ADF t-stat (Δy, n, p=0)	-4.692	-2.604	-1.946	-1.613	Reject H_0

Table 1. ADF test results for log(RGNP), 1909–1970.

Interpretation:

Both the ADF t-statistic (–2.994) and the φ_3 F-statistic (4.619) are above their 5% critical values of –3.49 and 6.49 respectively. We therefore **fail to reject the null hypothesis of a unit root** at any conventional significance level. The φ_3 test also confirms that no significant deterministic trend is present beyond the stochastic one.

In contrast, the ADF test on the **first difference** Δy_t gives –4.692, well below the 1% critical value of –2.604. The first difference is strongly stationary.

Conclusion: log(RGNP) is an I(1) process — a difference-stationary series driven by a stochastic trend. This replicates Nelson and Plosser (1982): most US macro series are indistinguishable from unit-root processes, implying that shocks have permanent effects on output.

B.1 Computer Exercise: Random Walk Monte Carlo Simulation

Setup. We simulate N = 100,000 independent sequences of the pure random walk

$$x_t = x_{t-1} + u_t, \quad u_t \sim N(0,1), \quad x_0 = 0, \quad T = 81$$

For each simulated series, three OLS models are estimated:

Model	Specification	Deterministics
1	$x_t = \rho\, x_{t-1} + u_t$	None
2	$x_t = c + \rho\, x_{t-1} + u_t$	Constant
3	$x_t = c + \delta t + \rho\, x_{t-1} + u_t$	Constant + Trend

(a) Percentiles of rho-hat and t-statistics

	Mean rho-hat	Std rho-hat	14th pct	86th pct
Model 1 (no constant)	0.9787	0.0380	0.9442	1.0091
Model 2 (constant)	0.9359	0.0525	0.8831	0.9847
Model 3 (const+trend)	0.8794	0.0681	0.8076	0.9465

Table 2. Distribution of OLS estimates of rho under the unit-root DGP.

	Mean t-stat	Std t-stat	14th pct	86th pct
Model 1 (no constant)	-0.421	0.990	-1.442	+0.672
Model 2 (constant)	-1.550	0.867	-2.448	-0.625
Model 3 (const+trend)	-2.219	0.792	-3.046	-1.403

Table 3. Distribution of t-statistics (rho-hat - 1) / se(rho-hat) under H_0: rho = 1.

(b) Distributions of rho-hat and t-statistics

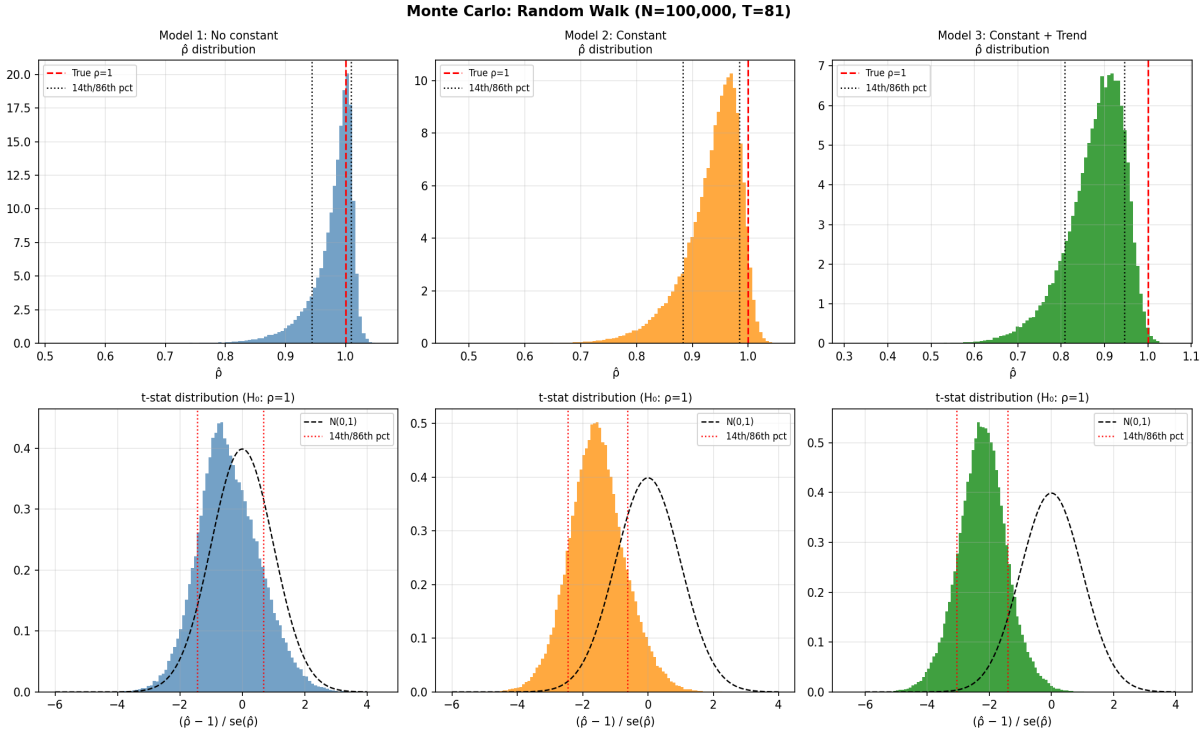

Figure 2. Top row: distribution of rho-hat for each model (red dashed = true rho = 1, dotted = 14th/86th percentiles). Bottom row: t-statistic distribution vs N(0,1) reference.

(c) Comments

Downward bias of rho-hat. Even though the true value is rho = 1, the OLS estimator is biased downward. The bias worsens as more deterministic components are included: Model 1 has mean rho-hat ≈ 0.979, while Model 3 has mean rho-hat ≈ 0.879. Including a constant or trend absorbs variation from x_{t-1}, pulling the estimated coefficient toward zero.

Non-standard t-distribution. The t-statistics are *not* normally distributed under the unit-root null. They are left-skewed and shifted well below zero — Model 3 has a mean t-stat of –2.22, far from the N(0,1) mean of zero. This is the Dickey-Fuller distribution.

Critical implication for inference. Using standard t-table critical values (–1.645 at 5%) would massively over-reject the unit-root null. The correct 5% left-tail cutoff is –1.44 (Model 1), –2.45 (Model 2), and –3.05 (Model 3) — all more negative than standard critical values.

Role of deterministics. Adding a constant or trend shifts the entire distribution of both rho-hat and the t-statistic further left. This is why Dickey-Fuller tables have three separate sets of critical values depending on the model specification.

B.2 Computer Exercise: BN Decomposition and HP Filter (US Real GDP)

Data. US Real GDP (GDPC1), quarterly, 1947:Q1–2019:Q4 (T = 292 observations, truncated at 2019:Q4 to exclude COVID). Log series: $y_t = \ln(GDP_t)$.

(a) ARMA(1,0) Fit and Beveridge-Nelson Decomposition

An ARMA(1,0) model with intercept is fitted to the first differences Δy_t by maximum likelihood. BIC also selects ARMA(1,0) from models up to ARMA(2,1), confirming the specification.

$$\text{Estimated model:} \quad \Delta y_t = 0.00775 + 0.3585\, \Delta y_{t-1} + \varepsilon_t, \quad \text{sigma}^2 = 7.47 \times 10^{-5}$$

Parameter	Estimate	Std. Error	z-stat	p-value	95% CI
c (intercept)	0.00775	0.00080	9.67	0.000	[0.006, 0.009]
φ (AR coef)	0.3585	0.047	7.66	0.000	[0.267, 0.450]
σ^2	7.47e-5	4.4e-6	16.97	0.000	[6.6, 8.3]e-5

Table 4. ARMA(1,0) estimation results for Δy_t, 1947:Q1–2019:Q4.

The long-run mean of Δy_t is mu-hat = c-hat / (1 – phi-hat) = 0.00775 / (1 – 0.3585) = **0.01207** per quarter (≈ 4.8% annualised).

Beveridge-Nelson decomposition. For ARMA(1,0) on Δy_t, the BN transitory component has a closed-form expression:

$$C^{BN}_t = -\,\text{phi-hat} / (1 - \text{phi-hat}) \times (\Delta y_t - \text{mu-hat})$$

$$\text{With phi-hat} = 0.3585: \quad C^{BN}_t = -0.558 \times (\Delta y_t - 0.01207)$$

The BN permanent component is $\text{tau}^{BN}_t = y_t - C^{BN}_t$. The cycle captures how much current growth deviates from the long-run average, scaled by the persistence of those deviations.

(b) Hodrick-Prescott Filter

The HP filter finds the trend tau^{HP}_t by minimising:

$$\Sigma(y_t - \text{tau}_t)^2 + \lambda\, \Sigma(\Delta^2 \text{tau}_t)^2$$

with λ **= 1,600** (standard for quarterly data). A higher λ produces a smoother trend.

(c) Comparison of Transitory Components

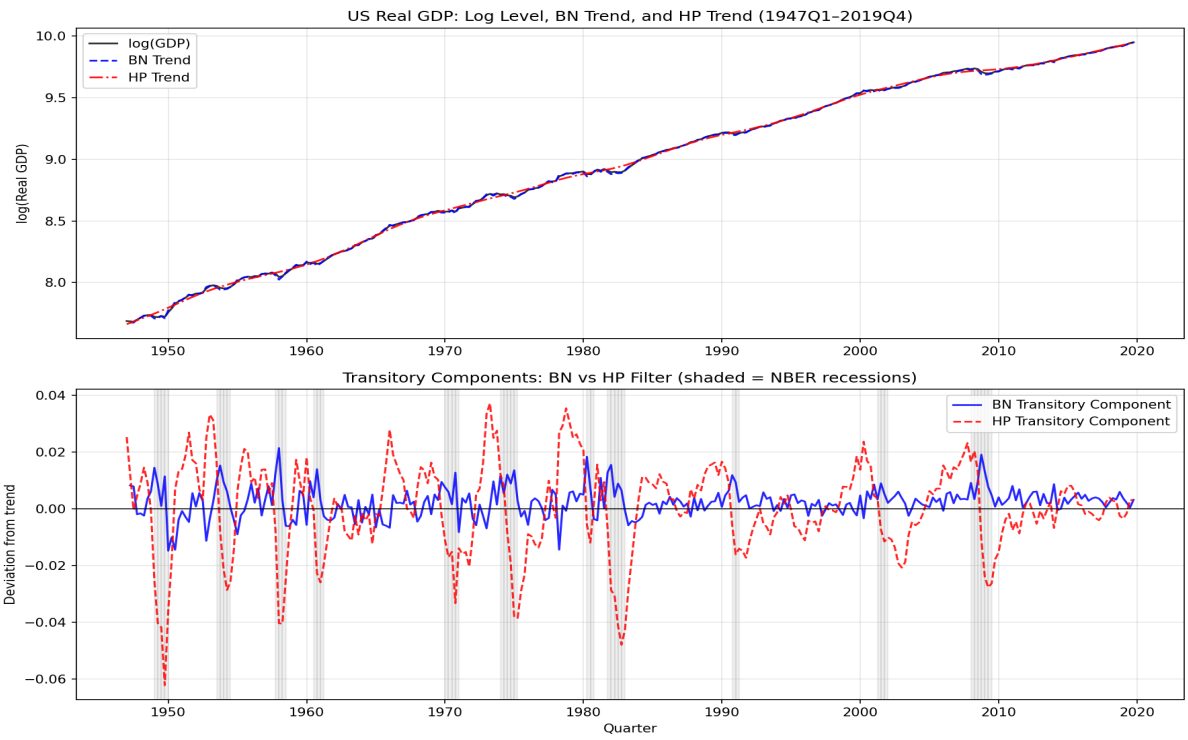

Figure 3. Top: log(GDP), BN trend and HP trend. Bottom: BN cycle (blue) vs HP cycle (red dashed), shaded = NBER recessions.

Statistic	BN Cycle	HP Cycle
Std. Deviation	0.0052	0.0158
Minimum	-0.0148	-0.0623
Maximum	+0.0214	+0.0372
Correlation (BN, HP)	-0.257	—

Table 5. Summary statistics for cyclical components.

Amplitude. The HP cycle has a standard deviation of 1.58% — about three times larger than the BN cycle (0.52%). The HP filter attributes large, persistent swings in GDP (e.g., the 2008–09 recession, reaching –6.2%) to the cycle. The BN cycle is smaller because positive autocorrelation in growth (phi = 0.36) means above-average growth today raises the permanent component by more than just the current increment, absorbing much of the swing.

Negative correlation. The two cycles are negatively correlated (–0.26), meaning they often move in opposite directions. The BN and HP decompositions embed fundamentally different views of what constitutes the "trend": the HP trend is a smooth curve, while the BN trend is a martingale that jumps every period.

Recession behaviour. Both cycles turn negative during NBER recessions (shaded grey). The HP cycle more clearly identifies business cycle turning points. The BN cycle is noisier at quarterly frequency with only an AR(1), but is theoretically grounded in rational expectations.

Conclusion: Both filters agree on the direction of cyclical swings around recessions but disagree on their magnitude. The choice depends on whether one views the trend as a smooth path (HP) or a stochastic process incorporating all predictable future growth (BN). Neither is universally correct.